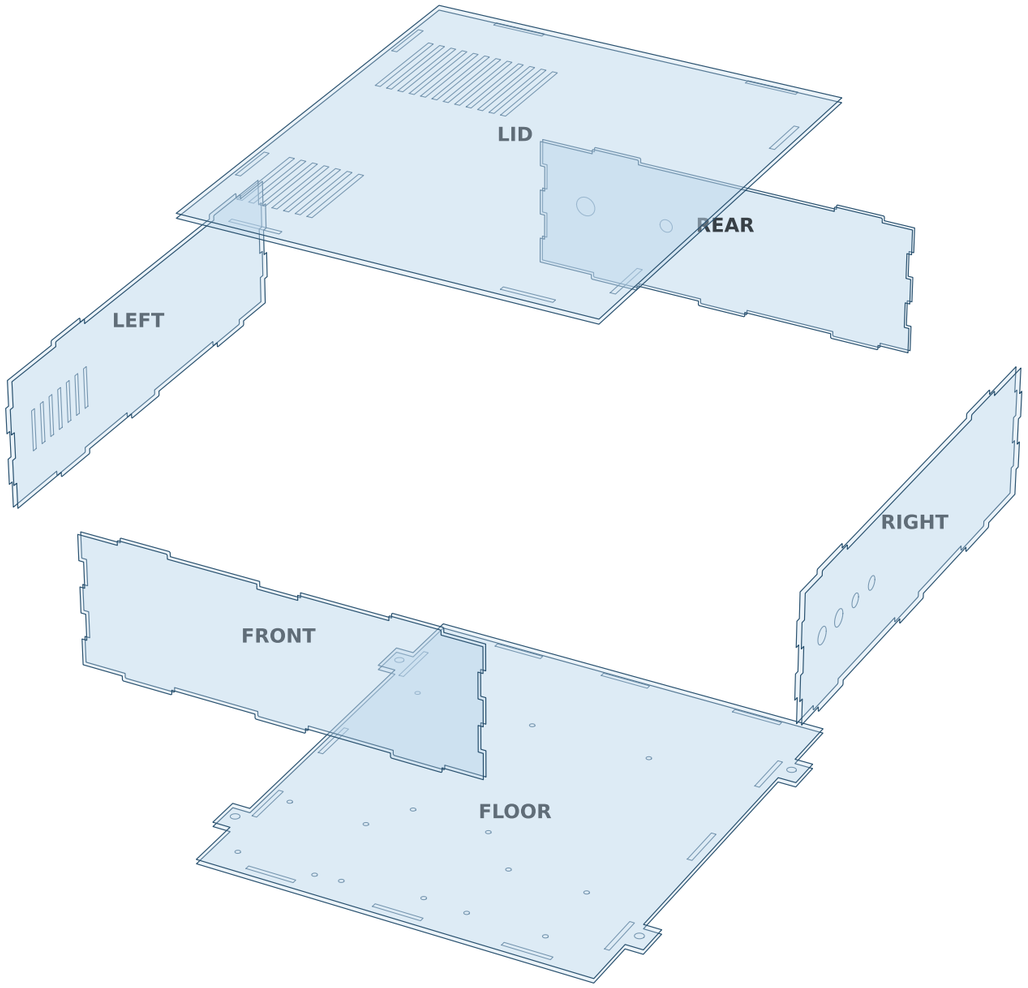
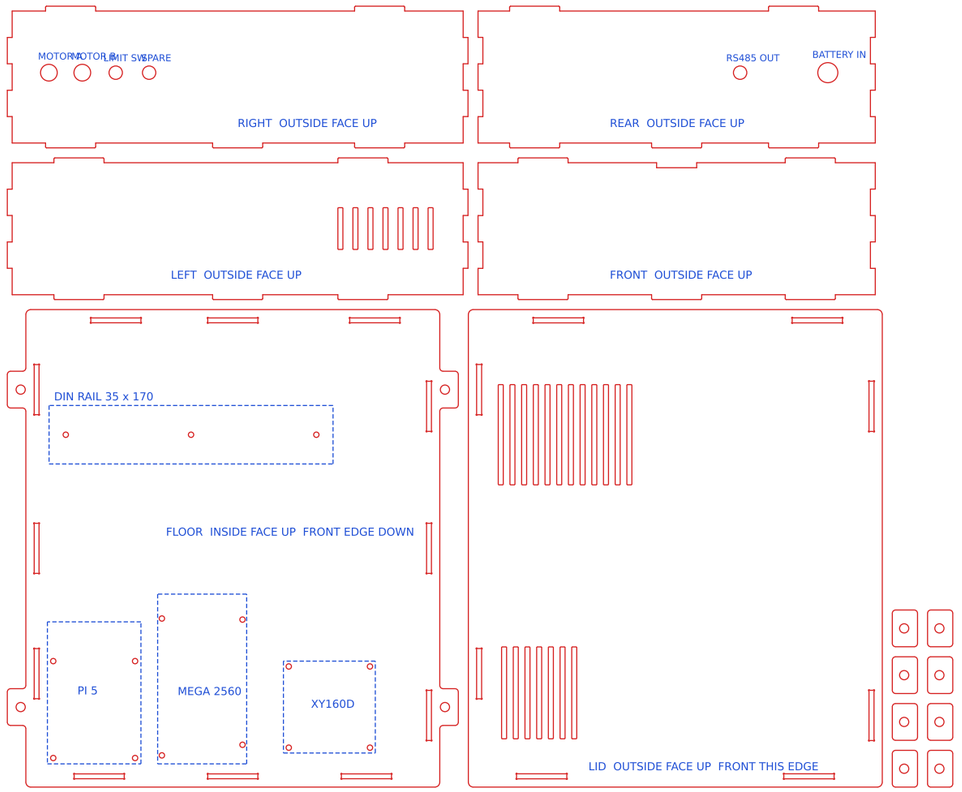
ASI-301 ELECTRONICS CASE V3 - EXPLODED VIEW AND SHEET LAYOUT





ASSEMBLY

1. FIT COUPON DONE, RESULT APPLIED (LAST PAGE). CUT THE PANELS FROM THE SAME SHEET.
2. KEEP FILM ON UNTIL ASSEMBLY. DRY-FIT EVERYTHING BEFORE ANY CEMENT.
3. JOIN THE FOUR WALLS INTO A RING AT THE CORNER FINGERS. TAB PATTERN ONLY ALLOWS ONE ORIENTATION.
4. PRESS THE RING INTO THE FLOOR SLOTS. PRESS NEAR EACH TAB, EVENLY, ON A FLAT TABLE.
5. WICK THIN SOLVENT ACRYLIC CEMENT INTO THE CORNERS AND SLOTS. PRESS FIT ALONE IS NOT FOR VIBRATION.
6. CEMENT TWO PADS STACKED UNDER EACH M5 EAR, HOLES ALIGNED (M5 SCREW AS A GUIDE).
7. LID PRESSES ON AND LIFTS OFF. NO CEMENT.

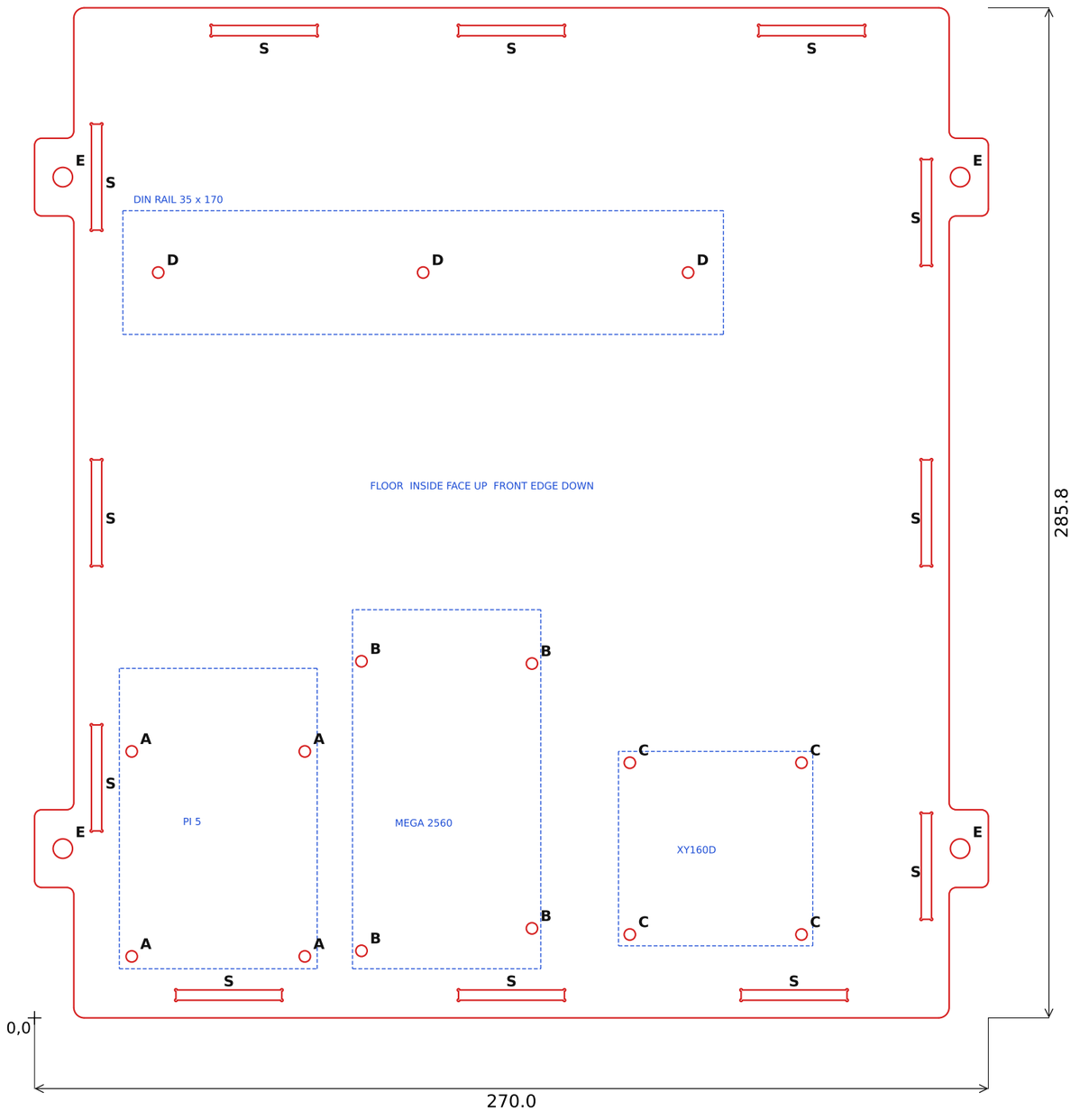
HARDWARE

12 x M3 MALE-FEMALE BRASS STANDOFF 10 + 6 mm (V2 HAD 11. 12TH = MEGA SUPPORT POST)
12 x M3 NUT + WASHER, UNDER THE FLOOR (STUD THROUGH THE FLOOR)
3 x M3 x 8 + WASHER + NUT, DIN RAIL. HEAD IN THE RAIL, NUT UNDER THE FLOOR
4 x M5 SCREW + WIDE WASHER, EARS. GRIP = 8.7 mm OF ACRYLIC. DO NOT OVERTIGHTEN

DO NOT USE ON ACRYLIC: THREADLOCKER, SUPERGLUE, ALCOHOL OR ACETONE. THEY CRACK LASER-CUT EDGES.
CUT LENGTH 10.2 m. BOX 237.8 x 275.8 x 84.8 (WALLS), 247.8 x 285.8 (FLOOR AND LID), 270 OVER EARS.

TITLE	ASI-301 ELECTRONICS CASE V3 (LASER-CUT ACRYLIC)
PART	OVERVIEW (REPLACES REV A, DO NOT CUT REV A)
MATERIAL	CLEAR ACRYLIC 3 mm NOM. (2.90 TESTED)
QTY	6 PANELS + 8 PADS
FINISH	AS CUT, FILM ON
DRAWN	JEON WONJE 2026-11-06 REV B
FILE	ASI301_case_acrylic_3mm_nest_revB.dxf

FLOOR (INSIDE FACE UP) - NTS - DATUM 0,0 BOTTOM-LEFT

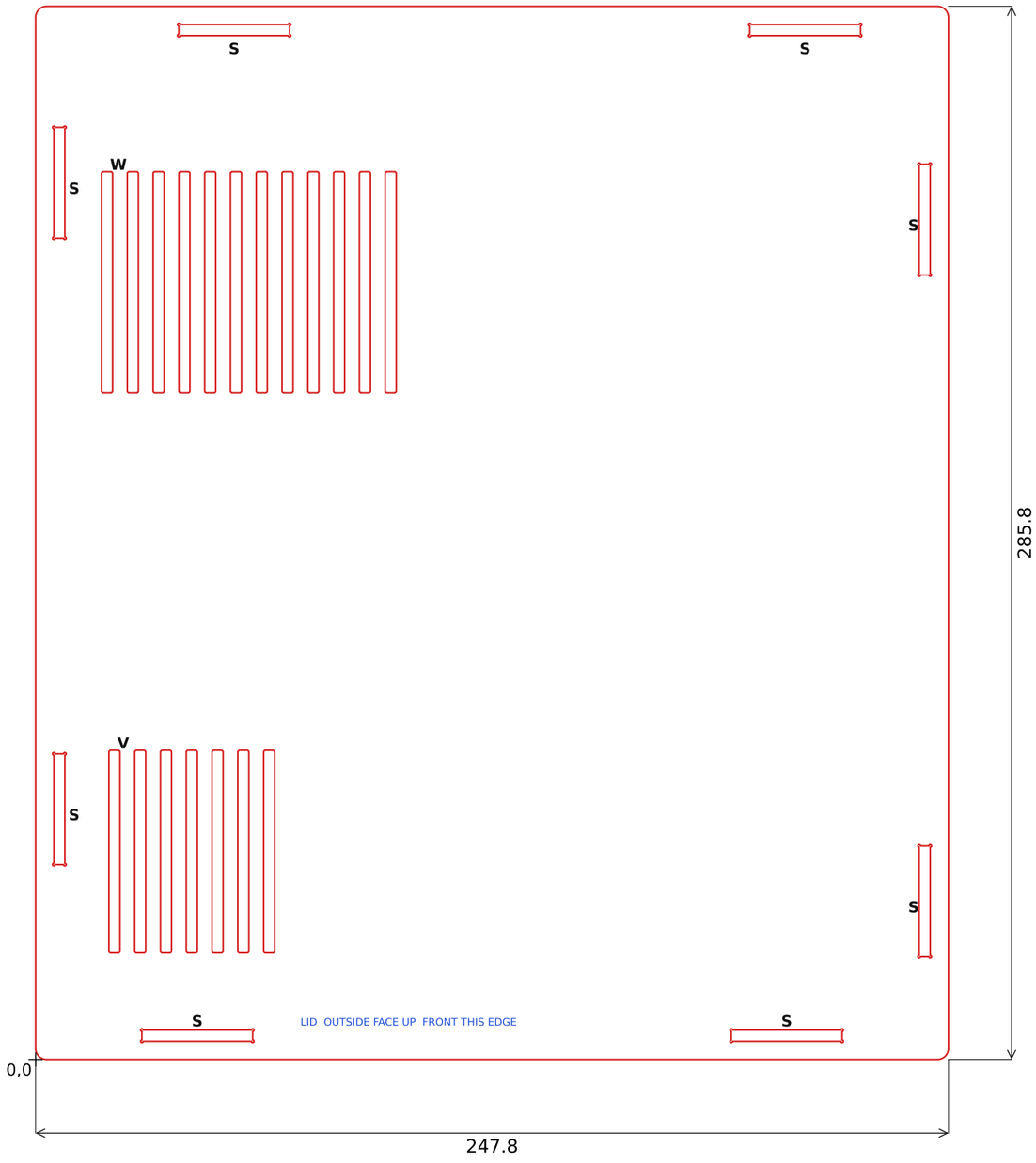


TAG	X	Y	SIZE	FEATURE
A	27.500	17.400	D3.2 THRU	PI 5 STANDOFF
A	27.500	75.400	D3.2 THRU	PI 5 STANDOFF
A	76.500	17.400	D3.2 THRU	PI 5 STANDOFF
A	76.500	75.400	D3.2 THRU	PI 5 STANDOFF
B	140.800	100.260	D3.2 THRU	MEGA STANDOFF
B	140.800	25.330	D3.2 THRU	MEGA STANDOFF
B	92.540	18.980	D3.2 THRU	MEGA STANDOFF
B	92.540	100.900	D3.2 THRU	MEGA SUPPORT STANDOFF
C	217.100	72.200	D3.2 THRU	XY160D STANDOFF
C	217.100	23.600	D3.2 THRU	XY160D STANDOFF
C	168.500	72.200	D3.2 THRU	XY160D STANDOFF
C	168.500	23.600	D3.2 THRU	XY160D STANDOFF
D	35.000	210.900	D3.2 THRU	DIN RAIL M3
D	110.000	210.900	D3.2 THRU	DIN RAIL M3
D	185.000	210.900	D3.2 THRU	DIN RAIL M3
E	8.000	47.900	D5.5 THRU	M5 MOUNT
E	8.000	237.900	D5.5 THRU	M5 MOUNT
E	262.000	47.900	D5.5 THRU	M5 MOUNT
E	262.000	237.900	D5.5 THRU	M5 MOUNT
S	55.000	6.450	29.975 x 2.900	SLOT FRONT
S	135.000	6.450	29.975 x 2.900	SLOT FRONT
S	215.000	6.450	29.975 x 2.900	SLOT FRONT
S	65.000	279.350	29.975 x 2.900	SLOT REAR
S	135.000	279.350	29.975 x 2.900	SLOT REAR
S	220.000	279.350	29.975 x 2.900	SLOT REAR
S	252.450	42.900	2.900 x 29.975	SLOT RIGHT
S	252.450	142.900	2.900 x 29.975	SLOT RIGHT
S	252.450	227.900	2.900 x 29.975	SLOT RIGHT
S	17.550	67.900	2.900 x 29.975	SLOT LEFT
S	17.550	142.900	2.900 x 29.975	SLOT LEFT
S	17.550	237.900	2.900 x 29.975	SLOT LEFT

1. MATERIAL: CLEAR ACRYLIC (PMMA), 3 mm NOMINAL, 2.90 AS TESTED. CUT ALL PANELS FROM THE COUPON'S SHEET. FILM ON.
2. DXF LAYER CUT (RED) = CUT THROUGH. LAYER ENGRAVE (BLUE) = LIGHT SURFACE ENGRAVE, OPTIONAL.
3. DXF ALREADY OFFSET FOR 0.20 mm KERF: CUT ON THE LINE, NO TOOL OFFSET. THIS DRAWING SHOWS FINISHED SIZES.
4. ALL DIMENSIONS IN mm. X/Y = FEATURE CENTRE FROM DATUM 0,0 AT THE BOTTOM-LEFT CORNER. DXF GOVERNS, DO NOT SCALE.
5. SLOTS S ARE THE WALL PRESS-FIT, SIZED FROM THE FIT COUPON: FINISHED SIZE AS TABLED, CORNER RELIEF R0.35 (DRAWN).
6. ENGRAVE IS ON THE INSIDE FACE: BOARD AND DIN RAIL FOOTPRINTS. OUTSIDE CORNERS R3, EAR CORNERS R2.

TITLE	ASI-301 ELECTRONICS CASE V3 (LASER-CUT ACRYLIC)
PART	FLOOR (INSIDE FACE UP)
MATERIAL	CLEAR ACRYLIC 3 mm NOM. (2.90 TESTED)
QTY	1
FINISH	AS CUT, FILM ON
DRAWN	JEON WONJE 2026-11-06 REV B
FILE	ASI301_case_acrylic_3mm_nest_revB.dxf

LID (OUTSIDE FACE UP) - NTS - DATUM 0,0 BOTTOM-LEFT



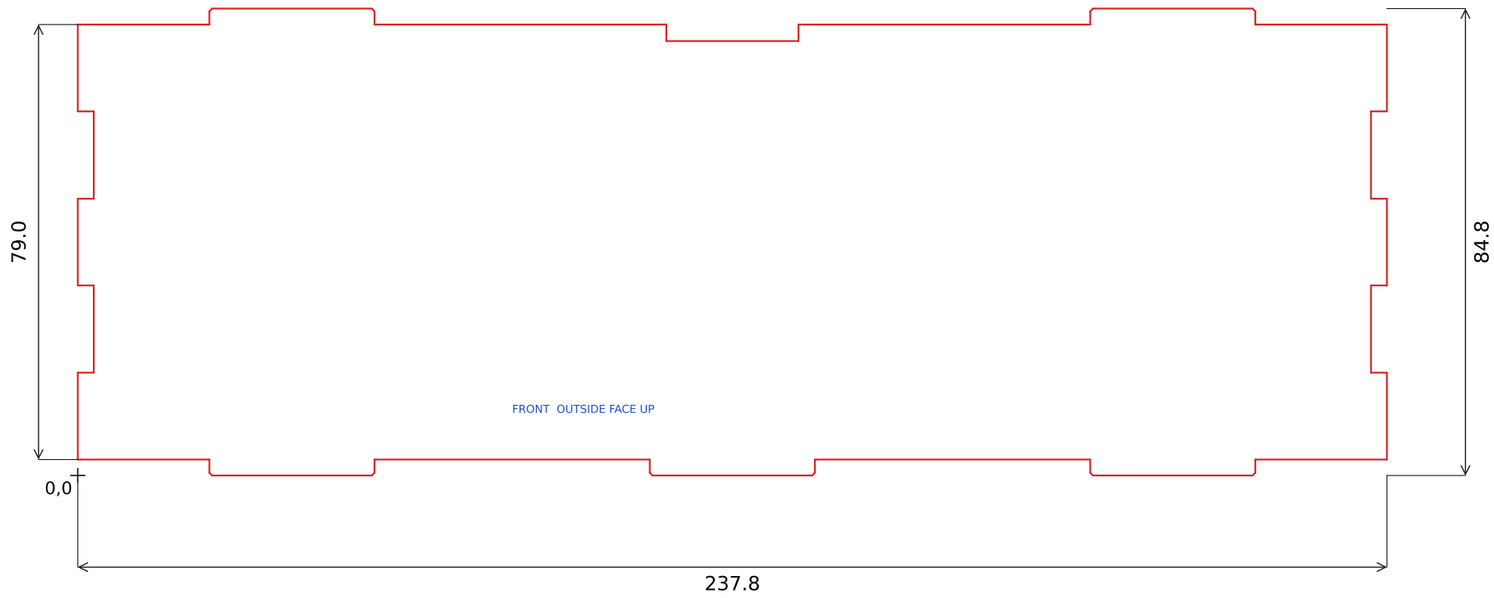
TAG	X	Y	SIZE	FEATURE
S	43.900	6.450	30.075 x 3.000	SLOT FRONT
S	203.900	6.450	30.075 x 3.000	SLOT FRONT
S	53.900	279.350	30.075 x 3.000	SLOT REAR
S	208.900	279.350	30.075 x 3.000	SLOT REAR
S	241.350	42.900	3.000 x 30.075	SLOT RIGHT
S	241.350	227.900	3.000 x 30.075	SLOT RIGHT
S	6.450	67.900	3.000 x 30.075	SLOT LEFT
S	6.450	237.900	3.000 x 30.075	SLOT LEFT
V	21.400	56.400	3.0 x 55.0 R0.5	VENT, PI 5 COOLER
V	28.400	56.400	3.0 x 55.0 R0.5	VENT, PI 5 COOLER
V	35.400	56.400	3.0 x 55.0 R0.5	VENT, PI 5 COOLER
V	42.400	56.400	3.0 x 55.0 R0.5	VENT, PI 5 COOLER
V	49.400	56.400	3.0 x 55.0 R0.5	VENT, PI 5 COOLER
V	56.400	56.400	3.0 x 55.0 R0.5	VENT, PI 5 COOLER
V	63.400	56.400	3.0 x 55.0 R0.5	VENT, PI 5 COOLER
W	19.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	26.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	33.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	40.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	47.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	54.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	61.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	68.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	75.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	82.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	89.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE
W	96.400	210.900	3.0 x 60.0 R0.5	VENT, DDR + WAVESHARE

1. MATERIAL: CLEAR ACRYLIC (PMMA), 3 mm NOMINAL, 2.90 AS TESTED. CUT ALL PANELS FROM THE COUPON'S SHEET. FILM ON.
2. DXF LAYER CUT (RED) = CUT THROUGH. LAYER ENGRAVE (BLUE) = LIGHT SURFACE ENGRAVE, OPTIONAL.
3. DXF ALREADY OFFSET FOR 0.20 mm KERF: CUT ON THE LINE, NO TOOL OFFSET. THIS DRAWING SHOWS FINISHED SIZES.
4. ALL DIMENSIONS IN mm. X/Y = FEATURE CENTRE FROM DATUM 0,0 AT THE BOTTOM-LEFT CORNER. DXF GOVERNS, DO NOT SCALE.
5. SLOTS S ARE 0.10 LARGER THAN THE FLOOR SLOTS: LID LIFTS OFF BY HAND. OUTSIDE CORNERS R3.

TITLE	ASI-301 ELECTRONICS CASE V3 (LASER-CUT ACRYLIC)
PART	LID (OUTSIDE FACE UP)
MATERIAL	CLEAR ACRYLIC 3 mm NOM. (2.90 TESTED)
QTY	1
FINISH	AS CUT, FILM ON
DRAWN	JEON WONJE 2026-11-06 REV B
FILE	ASI301_case_acrylic_3mm_nest_revB.dxf

FRONT WALL (OUTSIDE FACE UP) - NTS - DATUM 0,0 BOTTOM-LEFT

TAG	X	Y	SIZE	FEATURE
T	38.900	1.450	30.0 x 2.90	BOTTOM TAB
T	118.900	1.450	30.0 x 2.90	BOTTOM TAB
T	198.900	1.450	30.0 x 2.90	BOTTOM TAB
U	38.900	83.350	30.0 x 2.90	TOP TAB
U	198.900	83.350	30.0 x 2.90	TOP TAB

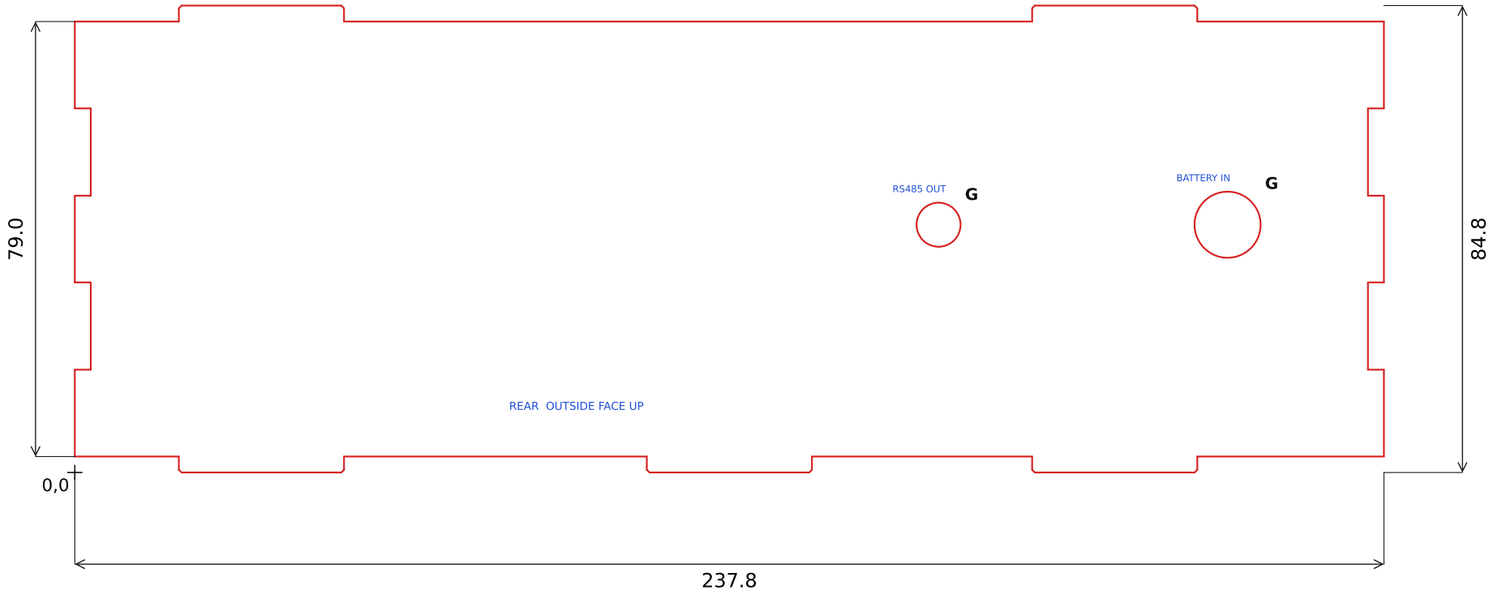


1. MATERIAL: CLEAR ACRYLIC (PMMA), 3 mm NOMINAL, 2.90 AS TESTED. CUT ALL PANELS FROM THE COUPON'S SHEET. FILM ON.
2. DXF LAYER CUT (RED) = CUT THROUGH. LAYER ENGRAVE (BLUE) = LIGHT SURFACE ENGRAVE, OPTIONAL.
3. DXF ALREADY OFFSET FOR 0.20 mm KERF: CUT ON THE LINE, NO TOOL OFFSET. THIS DRAWING SHOWS FINISHED SIZES.
4. ALL DIMENSIONS IN mm. X/Y = FEATURE CENTRE FROM DATUM 0,0 AT THE BOTTOM-LEFT CORNER. DXF GOVERNS, DO NOT SCALE.
5. TABS T PRESS INTO THE FLOOR, TABS U INTO THE LID. TAB TIP CHAMFER 0.5.
6. CORNERS: 5 ZONES OF 15.8, FINGER DEPTH 2.90, 0.05 CLEARANCE UP THE CORNER. DXF GOVERNS.

TITLE	ASI-301 ELECTRONICS CASE V3 (LASER-CUT ACRYLIC)
PART	FRONT WALL (OUTSIDE FACE UP)
MATERIAL	CLEAR ACRYLIC 3 mm NOM. (2.90 TESTED)
QTY	1
FINISH	AS CUT, FILM ON
DRAWN	JEON WONJE 2026-11-06 REV B
FILE	ASI301_case_acrylic_3mm_nest_revB.dxf

REAR WALL (OUTSIDE FACE UP) - NTS - DATUM 0,0 BOTTOM-LEFT

TAG	X	Y	SIZE	FEATURE
T	188.900	1.450	30.0 x 2.90	BOTTOM TAB
T	118.900	1.450	30.0 x 2.90	BOTTOM TAB
T	33.900	1.450	30.0 x 2.90	BOTTOM TAB
U	188.900	83.350	30.0 x 2.90	TOP TAB
U	33.900	83.350	30.0 x 2.90	TOP TAB
G	209.400	45.000	D12.0 THRU	BATTERY IN
G	156.900	45.000	D8.0 THRU	RS485 OUT

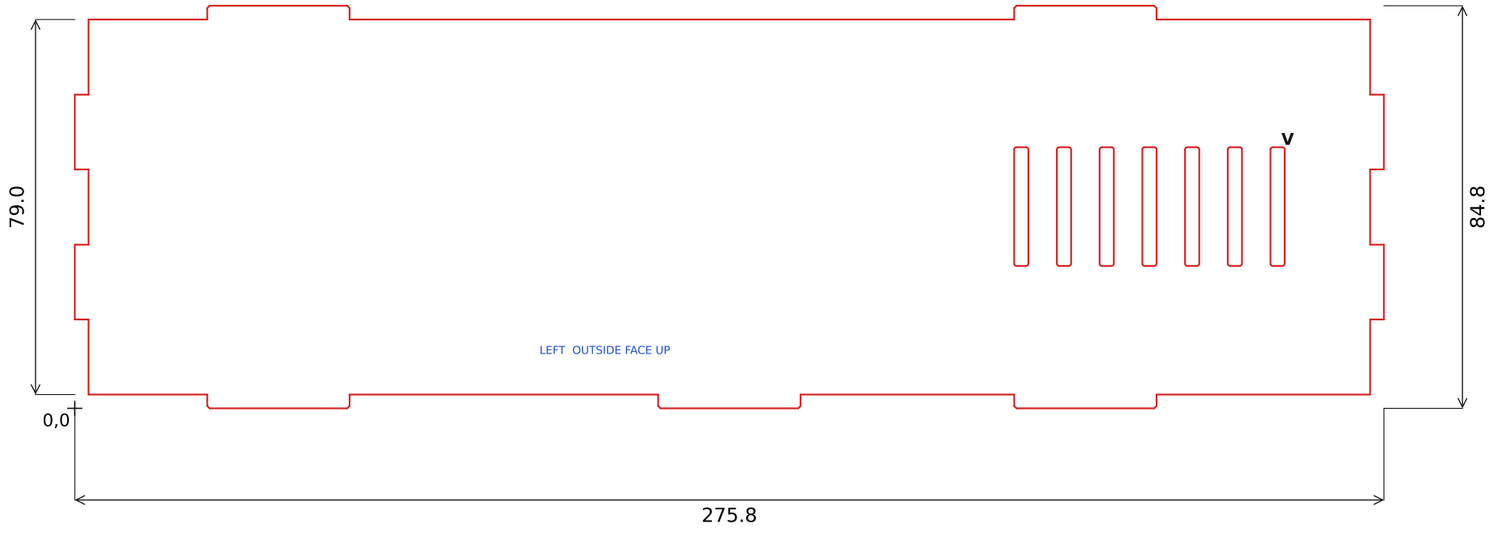


1. MATERIAL: CLEAR ACRYLIC (PMMA), 3 mm NOMINAL, 2.90 AS TESTED. CUT ALL PANELS FROM THE COUPON'S SHEET. FILM ON.
2. DXF LAYER CUT (RED) = CUT THROUGH. LAYER ENGRAVE (BLUE) = LIGHT SURFACE ENGRAVE, OPTIONAL.
3. DXF ALREADY OFFSET FOR 0.20 mm KERF: CUT ON THE LINE, NO TOOL OFFSET. THIS DRAWING SHOWS FINISHED SIZES.
4. ALL DIMENSIONS IN mm. X/Y = FEATURE CENTRE FROM DATUM 0,0 AT THE BOTTOM-LEFT CORNER. DXF GOVERNS, DO NOT SCALE.
5. TABS T PRESS INTO THE FLOOR, TABS U INTO THE LID. TAB TIP CHAMFER 0.5.
6. CORNERS: 5 ZONES OF 15.8, FINGER DEPTH 2.90, 0.05 CLEARANCE UP THE CORNER. DXF GOVERNS.

TITLE	ASI-301 ELECTRONICS CASE V3 (LASER-CUT ACRYLIC)
PART	REAR WALL (OUTSIDE FACE UP)
MATERIAL	CLEAR ACRYLIC 3 mm NOM. (2.90 TESTED)
QTY	1
FINISH	AS CUT, FILM ON
DRAWN	JEON WONJE 2026-11-06 REV B
FILE	ASI301_case_acrylic_3mm_nest_revB.dxf

LEFT WALL (OUTSIDE FACE UP) - NTS - DATUM 0,0 BOTTOM-LEFT

TAG	X	Y	SIZE	FEATURE
T	212.900	1.450	30.0 x 2.90	BOTTOM TAB
T	137.900	1.450	30.0 x 2.90	BOTTOM TAB
T	42.900	1.450	30.0 x 2.90	BOTTOM TAB
U	212.900	83.350	30.0 x 2.90	TOP TAB
U	42.900	83.350	30.0 x 2.90	TOP TAB
V	253.400	42.500	3.0 x 25.0 R0.5	VENT, PI 5
V	244.400	42.500	3.0 x 25.0 R0.5	VENT, PI 5
V	235.400	42.500	3.0 x 25.0 R0.5	VENT, PI 5
V	226.400	42.500	3.0 x 25.0 R0.5	VENT, PI 5
V	217.400	42.500	3.0 x 25.0 R0.5	VENT, PI 5
V	208.400	42.500	3.0 x 25.0 R0.5	VENT, PI 5
V	199.400	42.500	3.0 x 25.0 R0.5	VENT, PI 5

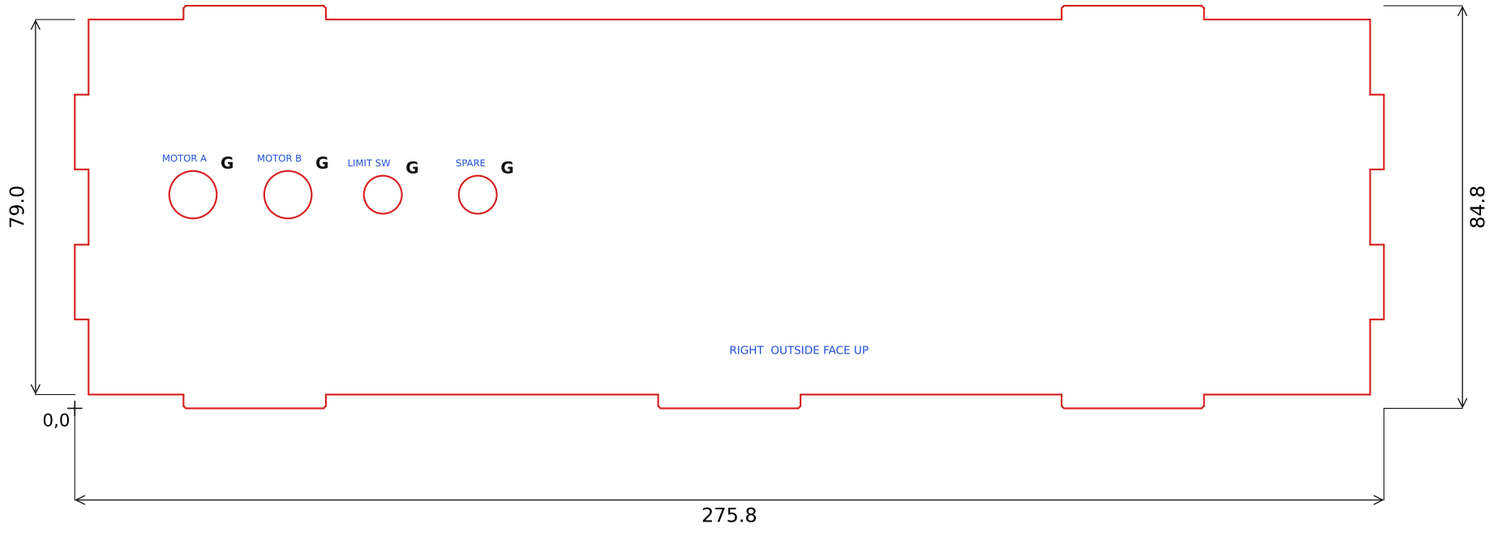


1. MATERIAL: CLEAR ACRYLIC (PMMA), 3 mm NOMINAL, 2.90 AS TESTED. CUT ALL PANELS FROM THE COUPON'S SHEET. FILM ON.
2. DXF LAYER CUT (RED) = CUT THROUGH. LAYER ENGRAVE (BLUE) = LIGHT SURFACE ENGRAVE, OPTIONAL.
3. DXF ALREADY OFFSET FOR 0.20 mm KERF: CUT ON THE LINE, NO TOOL OFFSET. THIS DRAWING SHOWS FINISHED SIZES.
4. ALL DIMENSIONS IN mm. X/Y = FEATURE CENTRE FROM DATUM 0,0 AT THE BOTTOM-LEFT CORNER. DXF GOVERNS, DO NOT SCALE.
5. TABS T PRESS INTO THE FLOOR, TABS U INTO THE LID. TAB TIP CHAMFER 0.5.
6. CORNERS: 5 ZONES OF 15.8, FINGER DEPTH 2.90, 0.05 CLEARANCE UP THE CORNER. DXF GOVERNS.

TITLE	ASI-301 ELECTRONICS CASE V3 (LASER-CUT ACRYLIC)
PART	LEFT WALL (OUTSIDE FACE UP)
MATERIAL	CLEAR ACRYLIC 3 mm NOM. (2.90 TESTED)
QTY	1
FINISH	AS CUT, FILM ON
DRAWN	JEON WONJE 2026-11-06 REV B
FILE	ASI301_case_acrylic_3mm_nest_revB.dxf

RIGHT WALL (OUTSIDE FACE UP) - NTS - DATUM 0,0 BOTTOM-LEFT

TAG	X	Y	SIZE	FEATURE
T	37.900	1.450	30.0 x 2.90	BOTTOM TAB
T	137.900	1.450	30.0 x 2.90	BOTTOM TAB
T	222.900	1.450	30.0 x 2.90	BOTTOM TAB
U	37.900	83.350	30.0 x 2.90	TOP TAB
U	222.900	83.350	30.0 x 2.90	TOP TAB
G	24.900	45.000	D10.0 THRU	MOTOR A
G	44.900	45.000	D10.0 THRU	MOTOR B
G	64.900	45.000	D8.0 THRU	LIMIT SW
G	84.900	45.000	D8.0 THRU	SPARE

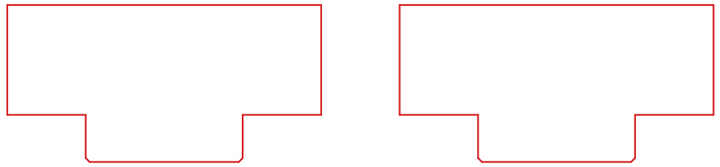


1. MATERIAL: CLEAR ACRYLIC (PMMA), 3 mm NOMINAL, 2.90 AS TESTED. CUT ALL PANELS FROM THE COUPON'S SHEET. FILM ON.
2. DXF LAYER CUT (RED) = CUT THROUGH. LAYER ENGRAVE (BLUE) = LIGHT SURFACE ENGRAVE, OPTIONAL.
3. DXF ALREADY OFFSET FOR 0.20 mm KERF: CUT ON THE LINE, NO TOOL OFFSET. THIS DRAWING SHOWS FINISHED SIZES.
4. ALL DIMENSIONS IN mm. X/Y = FEATURE CENTRE FROM DATUM 0,0 AT THE BOTTOM-LEFT CORNER. DXF GOVERNS, DO NOT SCALE.
5. TABS T PRESS INTO THE FLOOR, TABS U INTO THE LID. TAB TIP CHAMFER 0.5.
6. CORNERS: 5 ZONES OF 15.8, FINGER DEPTH 2.90, 0.05 CLEARANCE UP THE CORNER. DXF GOVERNS.

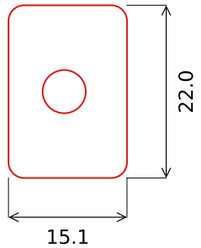
TITLE	ASI-301 ELECTRONICS CASE V3 (LASER-CUT ACRYLIC)
PART	RIGHT WALL (OUTSIDE FACE UP)
MATERIAL	CLEAR ACRYLIC 3 mm NOM. (2.90 TESTED)
QTY	1
FINISH	AS CUT, FILM ON
DRAWN	JEON WONJE 2026-11-06 REV B
FILE	ASI301_case_acrylic_3mm_nest_revB.dxf

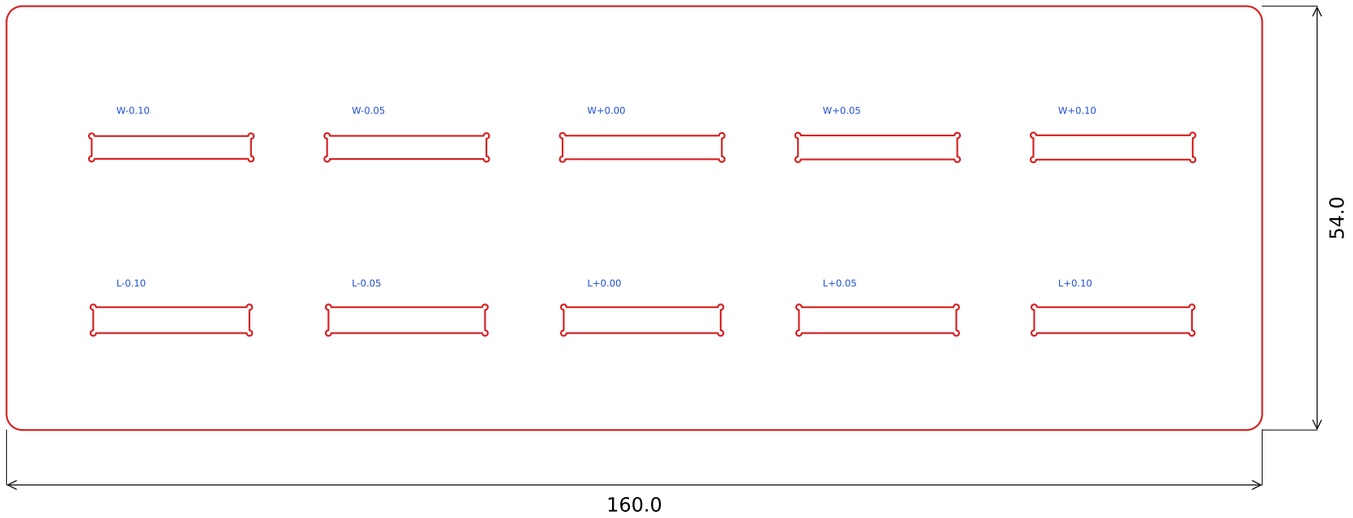
FIT COUPON AND M5 PAD - NTS

TEST TABS, QTY 2: 20.0 x 3.0 TAB



PAD QTY 8: D5.5, 7.1 FROM INNER END





COUPON RESULT, 6 NOV 2026

Best slots: L-0.05 to L+0.00, and W-0.10.

Applied in Rev B:
- sheet 2.90, kerf 0.20
- floor slots 2.90 x 29.975
- lid slots 3.00 x 30.075

Coupon drawn below as cut, at 3.00 nominal.
Re-test on a new sheet: re-run the script with
--sheet <t> --press-w <W> --press-l <L>

Acrylic does not give. A fit tighter than a firm hand press cracks during assembly or later.

1. MATERIAL: CLEAR ACRYLIC (PMMA), 3 mm NOMINAL, 2.90 AS TESTED. CUT ALL PANELS FROM THE COUPON'S SHEET. FILM ON.
2. DXF LAYER CUT (RED) = CUT THROUGH. LAYER ENGRAVE (BLUE) = LIGHT SURFACE ENGRAVE, OPTIONAL.
3. DXF ALREADY OFFSET FOR 0.20 mm KERF: CUT ON THE LINE, NO TOOL OFFSET. THIS DRAWING SHOWS FINISHED SIZES.
4. ALL DIMENSIONS IN mm. X/Y = FEATURE CENTRE FROM DATUM 0,0 AT THE BOTTOM-LEFT CORNER. DXF GOVERNS, DO NOT SCALE.

TITLE	ASI-301 ELECTRONICS CASE V3 (LASER-CUT ACRYLIC)
PART	FIT COUPON, TEST TAB, PAD
MATERIAL	CLEAR ACRYLIC 3 mm NOM. (2.90 TESTED)
QTY	COUPON 1, TAB 2, PAD 8
FINISH	AS CUT, FILM ON
DRAWN	JEON WONJE 2026-11-06 REV B
FILE	ASI301_fit_coupon_3mm.dxf